

Preetham Raj

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Professional Summary

AI/ML Engineer with 5+ years of experience building production-grade machine learning systems and NLP pipelines. Expert in Python, FastAPI, LangChain, and cloud infrastructure (AWS/GCP). Delivered 3 enterprise chatbot systems serving 50k+ daily users. Strong background in distributed systems, PostgreSQL, Redis, and Docker/Kubernetes.

Technical Skills

Python | FastAPI | LangChain | LangGraph | PostgreSQL | Redis | Docker | Kubernetes | AWS (ECS, S3, Lambda, Bedrock) | GCP | PyTorch | TensorFlow | scikit-learn | pandas | NumPy | REST APIs | GraphQL | gRPC | CI/CD (GitHub Actions) | Vector Databases (pgvector, Pinecone) | RAG Systems | LLMops | Prompt Engineering | Agents (LangGraph/CrewAI)

Professional Experience

Senior AI/ML Engineer — TechCorp Innovations, Bengaluru, India

Jan 2023 – Present

- Architected and deployed a multi-agent RAG system using LangGraph and FastAPI, handling 50k+ daily queries with sub-500ms latency.
- Built an AI-powered job search platform with 15 LangGraph agents — resume optimization, cover letter generation, interview coaching, email outreach, and salary intelligence.
- Designed pgvector-based semantic search pipeline indexing 100k+ documents with HNSW indexing, achieving 95%+ recall@10.
- Led migration from monolithic Django to microservices architecture using FastAPI, reducing deployment time from 45min to 3min.
- Implemented comprehensive observability with OpenTelemetry, Grafana, and structured logging — reduced MTTR by 60%.

ML Engineer — DataFlow Systems, Hyderabad, India

Jun 2021 – Dec 2022

- Developed NLP pipeline for document classification and entity extraction using transformer models (BERT, RoBERTa), achieving 92% F1 score.
- Built REST API microservices with FastAPI serving ML models at 10k RPM, with Redis caching reducing inference latency by 40%.
- Implemented CI/CD pipeline with GitHub Actions and Docker, automating model training, evaluation, and deployment to AWS ECS.
- Created real-time anomaly detection system processing 1M+ events/day with sub-second alerting using Redis Streams.

Junior Data Scientist — StartupML, Remote, India

Jul 2019 – May 2021

- Built customer churn prediction model using XGBoost and feature engineering, improving retention by 18% and saving \$2M annually.
- Developed interactive dashboards with Streamlit for stakeholder visualization of ML model performance metrics and business KPIs.
- Automated ETL pipelines processing 500GB+ daily using Python, Apache Airflow, and PostgreSQL — reduced manual data processing by 85%.

Education

B.Tech in Computer Science & Engineering — Indian Institute of Technology (IIT), Madras

2019 | GPA: 8.7/10

Certifications

- AWS Solutions Architect Associate
- Google Cloud Professional ML Engineer
- Deep Learning Specialization — deeplearning.ai (Coursera)

Projects

CareerCraft AI Platform (Python, FastAPI, LangGraph, pgvector, Supabase)

End-to-end job search automation platform with 15 AI agents. Features: resume optimization, cover letter generation, interview coaching, Gmail integration, company research, salary intelligence, and one-click job application via browser automation.

Multi-Agent Customer Support Bot (LangChain, LangGraph, FastAPI, PostgreSQL, Redis)

Designed a supervisor-worker agent system with 8 specialized agents handling billing, technical support, and account management. Reduced support ticket resolution time by 65%.